Interpretable Card-Play Strategies by Evolving Weight-Agnostic Logic Networks: A Neurosymbolic Agent for Sueca

André Cardoso[1]

Departamento de Eletrónica, Telecomunicações e Informática, Universidade de Aveiro, Portugal

Abstract. We present a neurosymbolic agent for **Sueca**, a four-player partnership trick-taking card game of imperfect information. The agent is a **Weight-Agnostic Neural Network** (WANN; Gaier & Ha, 2019) whose nodes are logic gates (aggregations SUM, AND, OR; activations IDENTITY, NOT, THRESHOLD) and whose connections carry only a sign (± 1) under a shared weight sweep. The network reads a **35-feature, public-information-only belief state** and emits one of **three abstract play intents**, which a styled heuristic resolver maps to a legal card. Training is two-phase: supervised bootstrap on a Perfect-Information Monte Carlo (PIMC) expert dataset using PFS-NEAT starting from zero connections, then co-evolutionary self-play of separate **lead** and **follow** brains. A key contribution is a cheap rollout teacher that is **stronger than our hand-crafted heuristic baseline** (which we call *Elite*): it determinizes hidden cards and finishes each world with Elite playouts, reaching 62% against Elite while labeling the pretraining dataset $\sim 1000\times$ faster than a deep alpha-beta search. The champion **beats Elite on 52.1% $\pm$ 1.8% of deals ($n = 3000$)** and, crucially, compiles to a handful of **human-readable IF/THEN rules** via verified constant-folding and alias inlining: the follow brain reduces to 5 logic gates at depth 5. We give an honest analysis of the *resolver-floor* ceiling that limits realisable strength ($\approx$58% oracle cap) and the resulting interpretability-vs-strength trade-off.

Keywords: weight-agnostic neural networks · neuroevolution · neurosymbolic AI · rule extraction · imperfect-information games · trick-taking card games.

1 Introduction

Trick-taking card games are a classic AI-hard domain. Each deal confronts players with imperfect information (hidden hands), partnership coordination, and a combinatorially large state space. Sueca, the Portuguese national four-player game, adds distinctive structure: a 40-card deck with the counter-intuitive rank order A $>$ 7 $>$ K $>$ J $>$ Q $>$ 6 $>$ 5 $>$ 4 $>$ 3 $>$ 2 (the 7, or *manilha*, is second-highest), 120 total points per deal partitioned into game-point tiers, mandatory follow-suit with trump cuts, and public void tracking when a player fails to follow. These rules create a rich strategic texture: the decision to lead a trump to draw opponents' high cards, to preserve a side-suit ace for partner, or to duck a cheap trick and build a void for later cuts all matter.

Most strong card-playing AIs are black boxes. Deep PIMC solvers and neural policy networks achieve superhuman play in Bridge and Skat, but the resulting strategies are opaque matrices of floating-point weights. Our thesis is different: can we obtain *competitive* play that is simultaneously *human-auditable*, extractable as short IF/THEN rules that a player can read, verify, and learn from? The agent architecture we design to test this is, in one line:

Belief(35) → WANN of logic gates → Intent(3) → Styled Resolver → Legal Card.

Weight-Agnostic Neural Networks [4] are evolved topologies whose connections share a single weight value swept at evaluation. We deliberately restrict the node primitives to *logical gates* (AND, OR, NOT, threshold) rather than continuous activations precisely because discrete logic compiles to Boolean-ish rules. This neurosymbolic commitment, logic gates rather than learned weights, is what makes the compiled output *interpretable* rather than merely inspectable.

We benchmark against a ladder of three non-learned opponents of increasing strength: a uniform-random player (*RandomBot*); an earlier rule-based bot (*OldHeuristic*); and a strong hand-crafted heuristic we call *Elite*, which plays the cheapest card that wins the trick, cuts cheaply with trump when void, and otherwise discards its least valuable legal card. Elite is the bar to beat: it already defeats OldHeuristic roughly two times in three, and no prior version of our agent had matched it.

Two obstacles had kept every earlier version of this agent below the Elite bar, and naming them up front motivates our contributions. First, the abstract intent vocabulary *collapsed* during self-play: because the three intents were each individually weak, training drove the network toward one dominant intent while the others atrophied, leaving the best prior agent at only 30.2% against Elite. Second, the natural supervised teacher, a deep alpha-beta PIMC search, itself only *tied* Elite, so imitating it could never exceed Elite. We diagnose both in Sections 4 and 6; the four contributions below are the two fixes and what they revealed.

We make four contributions:

1. A neurosymbolic Sueca agent that **beats Elite** (52.1% $\pm$ 1.8%, $n = 3000$) while remaining interpretable: the first agent in this project to exceed Elite, up from the 30.2% of the collapsed predecessor described above.
2. A cheap **rollout policy-improvement teacher** achieving 62% vs. Elite ($\sim$1000$\times$ faster than deep alpha-beta), together with the diagnosis that it was the deep search's *myopic leaf evaluator*, not its depth, that capped earlier datasets at Elite-level imitation.

3. **Verified post-hoc rule minification** (constant-folding + alias inlining) that yields an iso-strength champion with **4.5× fewer logic gates** (29 vs. 132) and genuinely readable rules: the follow brain compiles to 5 gates at depth 5.
4. An **honest negative result**: the styled-resolver *floor* is the real strength ceiling (≈58% oracle cap for perfect intent selection); a stronger teacher buys *simplicity*, not strength, a finding that reframes the whole strength–interpretability narrative.

2 Background and Related Work

Our work draws on neuroevolution, neurosymbolic AI, and imperfect-information game playing.

Weight-Agnostic Neural Networks. Gaier and Ha [4] evolve network topologies whose connections share a single scalar weight swept at evaluation. We adapt WANNs in three ways: nodes use a *discrete palette* (IDENTITY, NOT, THRESHOLD activations; SUM, MIN/AND, MAX/OR aggregations); connections carry only a *sign* (±1) with no magnitude; and we sweep six shared weights $W \in \{-2.0, -1.0, -0.5, 0.5, 1.0, 2.0\}$, averaging fitness across the sweep. Sign-only edges at negative W express inhibitory rules (e.g. $1 - x$ via sign inversion), enriching the logic vocabulary without sacrificing interpretability.

NEAT and modular coevolution. NEAT [10] evolves topologies through speciation and complexification from minimal structures. PFS-NEAT [11] adds progressive feature selection: networks start from zero connections and structural mutations must demonstrate performance lift. We classify mutations as structural (triggers PFS) or non-structural (skips PFS) and use adaptive two-stage sampling (K=25 quick check; K=100 for borderline cases) with a FIFO tabu veto list [5]. Following L-NEAT [9], we partition the decision space into *lead* and *follow* brains, routing queries via the `Am_I_Leading` belief flag.

PIMC and evaluation. PIMC sampling [3] determinizes hidden cards, solves each perfect-information world, and aggregates outcomes. We use PIMC as a dataset teacher (best-of-three-intents by resolved-card EV) and for evaluation. For low variance we employ Common Random Numbers (CRN) [6]: every genome plays the identical deal/seat/opponent as the baseline bot, analogous to duplicate bridge scoring and the AIVAT framework [1].

Quality-diversity and parsimony. MAP-Elites [7] maintains a 10×10 grid of genomes by intent preference and aggression, sampled as Phase-1 opponents. Multi-objective parsimony [8,2] ranks genomes on a (performance, simplicity) Pareto front 50% of the time, controlling bloat.

Rule extraction. Most neurosymbolic rule extraction operates post-hoc on continuous-weight networks, producing approximations. By *constraining the network to logic gates during evolution*, our compiled rules are *exact*: the computation the network performs, not an approximation.

3 Sueca and Problem Formulation

3.1 Rules That Shape Strategy

Sueca uses a 40-card deck (standard deck minus 8s, 9s, 10s) with four players in fixed partnerships (seats 0&2 vs. 1&3) and counter-clockwise turn order. The rank order is **A** > **7** > **K** > **J** > **Q** > **6** > **5** > **4** > **3** > **2**, where the 7 (*manilha*) is second-highest. Points are A=11, 7=10, K=4, J=3, Q=2 (rest zero), totaling 120 per deal. The highest trump wins the trick; otherwise the highest of the led suit wins. Players *must* follow suit if able; failure reveals a void to all players. Game-point tiers (1 pt for 61–90, 2 pts for 91–119, 4 pts for 120) create non-linear scoring incentives.

3.2 POMDP Framing

Each seat perceives the game as a Partially Observable Markov Decision Process (POMDP). The true state (all four hands) is hidden; the observation consists of the player's own hand plus the public history of played cards, trick outcomes, and exposed voids. Crucially, our belief encoder **never leaks opponent hand data**: it uses only information a human player at the table can observe, as Section 3.3 makes concrete for the tactical features.

3.3 Belief State

The belief state comprises **35 features**, all normalized to $[0, 1]$. They fall into five categories:

- **Hand composition** (features 0–4): suit holdings, trump count, point density of remaining cards.
- **Trick context** (5–9): leading/following/last-to-play flags, partner-winning indicator, trick point value, whether the trick has been cut.
- **Void knowledge** (10–13): partner and opponent voids in led suit and trump, the public information that constrains future play.
- **Tactical affordances** (14–21): boss detection (holding the highest unplayed card in led suit or trump), "can I beat the current winner?", minimum winning and sacrifice costs. These features, added in a belief redesign, capture *what a player can do* rather than just what they hold.
- **Game progress** (22–34): remaining points, trick number, trumps remaining, score delta, void count, side-suit lengths and depletion, secured points, and meta-features counting known voids and depleted suits.

Why these are public. Every feature is a deterministic function of the player's own hand and the public play history, so none requires knowledge of an opponent's cards. *Boss* detection illustrates the point. A card is the "boss" of a suit when it is the highest-ranked card of that suit not yet seen in any completed or in-progress trick. The player computes this purely from cards everyone at the table can see, the pile of already-played cards together with its own hand; an opponent could hold a higher card of the suit only if such a card were still unseen, which is exactly what holding the boss rules out. The same reasoning covers "can I beat the current winner?", the cheapest winning and sacrifice costs, and the trump- and point-remaining counts: each compares the player's own legal cards against the publicly known set of played cards. Voids (features 10–13) are public for the same reason, since failing to follow suit is observed by every player at the table.

The full 35-feature layout is documented in the project README; we include it as an appendix for reference.

3.4 Three Intents and the Styled Resolver

The network outputs three continuous values, one per **play intent**. Argmax with random tie-breaking selects the active intent. The intents are not standalone card-selection heuristics; they are **styled deviations around a strong Elite policy**:

- **EFFICIENT_WIN** (intent 1): Delegates exactly to Elite: play the cheapest winner, else cheap cut, else cheapest legal sacrifice. This is the strong default.
- **MAX_FORCE** (intent 0): Elite + control dials. When trump-long, lead a low trump to *draw* opponents' trumps; otherwise play aggressively.
- **EQUITY_BUILDER** (intent 2): Elite + tempo dials. Lead the *shortest* side-suit to build a void; *duck* cheap tricks ($\leq$2 points) when not last to play; *preserve* trump rather than cut a cheap trick.

The **styled resolver** (`select_card_styled`) maps the chosen intent to a legal card contextually, guaranteeing 100% legality. Because every intent shares Elite's core and only adds a narrow deviation, each pure intent benchmarks at approximately Elite individually (48/47/46% vs. Elite): the "raised floor" that eliminates the Phase-1 policy-collapse basin and is the architectural enabler of the final win.

4 Method

Figure 1 shows the full pipeline from belief state to card. The left side (belief state, WANN, intents) is evolved; the right side (styled resolver, card) is a fixed heuristic. This section details each component.

4.1 WANN Representation

A genome is a directed graph. Connection genes are 5-tuples (innovation, source, destination, sign $\in \{\pm 1\}$, enabled). Node genes specify type (input, bias, hidden, output), activation (IDENTITY, NOT, THRESHOLD), and aggregation (SUM, MIN/AND, MAX/OR). The 39 base nodes are: 35 inputs, one bias (constant 1.0), and three outputs (one per intent). We exclude MEAN (float precision at the THRESHOLD boundary) and SIGMOID (breaks rule extraction). All outputs are clamped to $[0, 1]$; THRESHOLD fires at 0.5.

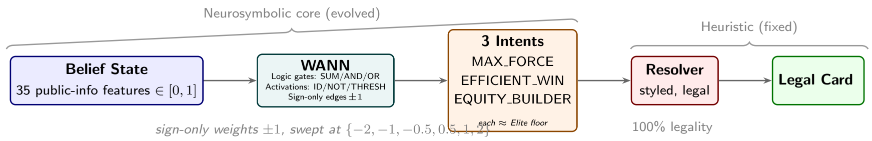

Fig. 1. The Sueca-WANN pipeline. A 35-feature public-information belief state feeds a weight-agnostic neural network built from logic gates (SUM/AND/OR aggregations; IDENTITY/NOT/THRESHOLD activations), which outputs three play intents. A fixed, hand-authored styled resolver maps the selected intent to a legal card. The neurosymbolic core (belief $\rightarrow$ WANN $\rightarrow$ intents) is evolved; the resolver is a deterministic heuristic guaranteeing 100% legality.

A connection from i to j carries only a sign: a positive edge passes out_i, while a negative edge *inverts* the signal to $1 - \mathrm{out}_i$ (logical NOT, since outputs lie in $[0, 1]$); the result is then scaled by the shared weight W before j aggregates it. At inference the network is evaluated at all six sweep weights $\{-2.0, -1.0, -0.5, 0.5, 1.0, 2.0\}$ and the per-intent outputs are averaged across the sweep before argmax intent selection.

4.2 Evolution

Both populations initialize with zero active connections. Structural mutations (add-node, add-connection) trigger PFS validation: the candidate is evaluated and retained only if it does not degrade accuracy. We use adaptive two-stage sampling (quick K=25, full K=100 for borderline cases) with a FIFO tabu veto list (capacity 1000). Non-structural mutations (toggle, flip-sign, change-activation, change-aggregation) skip PFS.

Genomes are speciated via compatibility distance with first-fit assignment (capped at 20 species). Rank-based selection precedes tournament selection. Multi-objective Pareto ranking on (performance, simplicity) operates 50% of the time; the other 50% uses performance-only ranking. A Hall of Fame (size 50) and a MAP-Elites 10×10 grid supply Phase-1 opponents. Fitness is the win-rate delta against HeuristicBot via CRN on duplicate deals with seat rotations; deals are re-seeded per generation (seed = gen) to prevent overfitting.

4.3 Two-Phase Curriculum

Phase 0 (generations 0–149): Supervised pretraining. The PIMC dataset is split by `Am_I_Leading` into separate lead/follow datasets. Fitness is classification accuracy; PFS-NEAT gates all structural mutations. The population of 1000 runs 150 generations (cleaner rollout-teacher labels need fewer gens than the earlier 200-gen default).

Phase 1 (generations 150–599): Co-evolutionary self-play. Lead and follow brains co-evolve: candidate lead WANNs pair with the reference follow

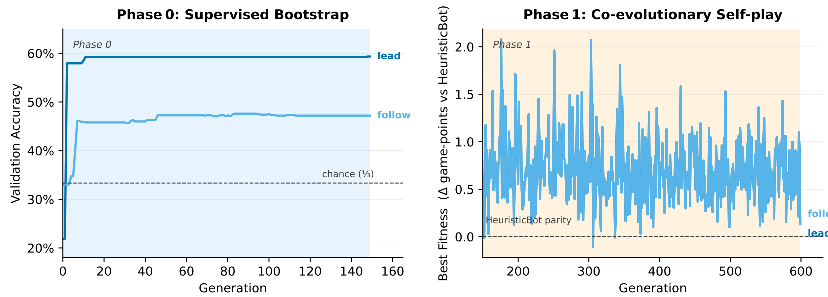

Fig. 2. Training curves for the final champion. Phase 0 (generations 0–149): supervised classification accuracy on lead and follow dataset splits, starting from zero connections. Phase 1 (generations 150–599): co-evolutionary self-play fitness (game-point delta vs. HeuristicBot). The Phase 0→1 transition re-evaluates HOF entries under Phase 1 fitness.

champion, and vice versa. The simulator dynamically routes each card decision to the lead or follow brain via `Am_I_Leading`. HOF entries are re-evaluated under Phase 1 fitness at the transition. Uniqueness filtering uses $O(1)$ innovation fingerprint hashing.

4.4 The Rollout Teacher

A critical finding emerged during diagnosis: the project's deep alpha-beta PIMC solver only *ties* Elite, despite searching to depth 4 with 200 worlds. The bottleneck was not the search depth but the **myopic leaf evaluation**: at the depth limit, alpha-beta returns the raw points-captured-so-far (`state.team_02_score`) with no positional estimate. Deeper search cannot fix a blind leaf.

We implemented a separate flat Monte-Carlo solver, `solve_pimc_rollout`, that avoids the search tree entirely. Per legal move, it determinizes the unseen cards, plays the candidate move, then finishes each world with **Elite in all four seats**. By the rollout policy-improvement theorem (a one-ply move followed by a base policy π is at least as strong as π itself), the resulting EV estimates are **supra-Elite**. Empirically, the rollout teacher achieves **62.0% vs. Elite** (where Elite self-play scores below 50% for the first team due to a fixed positional disadvantage). Critically, it is also $\sim$**1000**$\times$ **faster**: the rollout-teacher dataset of 15,000 labeled states generates in approximately 11 seconds versus hours for the alpha-beta pipeline.

Dataset labeling follows a best-of-three-intents protocol: each decisive state is labeled with the intent whose resolved card achieves the highest PIMC expected value. Statistically tied intents receive uniform multi-label weights; states where all three intents are tied (hence the intent choice is irrelevant) are rejected. This produces a natural, approximately binary EFFICIENT_WIN-vs-EQUITY_BUILDER split, with MAX_FORCE uniquely best in $\sim$1% of states (and 0% in the follow split). We generate with `soft-balance-min-ratio 0.0`

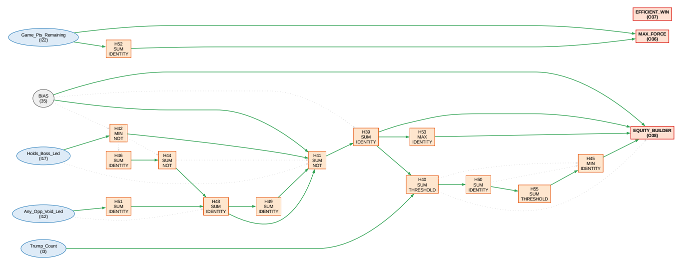

Fig. 3. The final champion's follow-brain topology before minification (17 hidden gates, 44 connections). After constant-folding and alias inlining, the active logic reduces to 5 gates at depth 5 (see Listing 1.1).

and disable inverse-frequency class weighting, since the rare MAX_FORCE bucket would receive disproportionate noise amplification.

5 Interpretability: From Network to Rules

The `compile-rules` tool walks the genome's DAG and emits one IF/THEN expression per output intent. Two **behavior-preserving rewrites** minimise the rules without altering the network's output:

1. **Constant-folding.** Nodes fed exclusively by the BIAS node (constant 1.0) or with empty aggregation sets are constants. Their values are computed via the exact `wann_network::forward` arithmetic and inlined wherever they appear. Constant nodes are dropped from the listing.
2. **Alias inlining.** A single-input IDENTITY node is a pure rename (under all three aggregations at W=1, SUM(x) = MIN(x) = MAX(x) = x). Chains of such aliases are flattened to their ultimate source, tracking negation parity (NOT∘NOT cancels). Single-operand AND and OR are similarly unwrapped. This rewrite is guarded to $W = 1$; at $W \neq 1$, the node computes clamp(src $\cdot$ W), which is not an alias.

Verification. A property test runs the real champion through `wann_network::forward` on 2000 random belief states at W=1 and asserts: (a) every folded constant is invariant to within 10^{-9}, and (b) every alias node equals its resolved source (negation parity applied) to within 10^{-9}. This means the dropped symbols provably carry no information the network uses: the reader can trust the rules are exact, not an approximation.

The minification is substantial. The follow brain before folding had 12 listed steps with a misleading "max depth" of 10 (seven of those steps were pure alias renames like `hidden_49 = hidden_48` and two were dead constants). After folding: **5 active logic gates at depth 5**. The lead brain reduces from 12 to 10 gates (2 aliases inlined) at depth 6.

Listing 1.1. Compiled and folded rules for the final champion's FOLLOW brain. MAX_FORCE is a one-feature game-phase toggle (2· Game_Pts_Remaining: force early, concede late). EFFICIENT_WIN is identically zero: the network never delegates to pure Elite. EQUITY_BUILDER gates on boss detection, opponent voids, trump count, and a threshold cascade. The LEAD brain (not shown) compiles to 10 gates at depth 6.

```
FOLLOW brain (compiled, folded):
  hidden_42 = NOT(Holds_Boss_Led)
  hidden_48 = (1 + Any_Opp_Void_Led)
  hidden_41 = NOT((1 + hidden_42 + hidden_48 + hidden_48))
  hidden_40 = THRESHOLD((hidden_41 + Trump_Count) > 0.5)
  hidden_55 = THRESHOLD(hidden_40 > 0.5)
  MAX_FORCE      = Game_Pts_Remaining + Game_Pts_Remaining
  EFFICIENT_WIN  = 0
  EQUITY_BUILDER = 1 + hidden_41 + hidden_55 + hidden_41
```

Fig. 4. Minified FOLLOW-brain rules. Before folding: 12 steps, max depth 10 (7 aliases, 2 dead constants, 1 single-operand AND). After: 5 active gates, max depth 5. The rule semantics are strategically sensible; see Section 7 for interpretation.

Table 1. Complexity metrics after constant-folding and alias inlining. Active gates count only information-carrying logic nodes.

Metric	**LEAD brain**	**FOLLOW brain**
Active hidden gates	10 (was 12)	**5** (was 12)
Live connections / enabled	20 / 22	13 / 27
Max logic depth	6	**5** (was 10)
Total input literals	10	5

Table 1 quantifies the post-folding complexity of both brains. The key metric is not total gates but *active* gates after removing constants, aliases, and single-operand wrappers, which represent the genuine, information-carrying logic.

6 Experiments and Results

6.1 Evaluation Protocol

All benchmarks use seat-rotated duplicate deals with CRN. We report win percentages with 95% CI half-widths (Bernoulli, normal approximation). The standard benchmark uses $n = 3000$ deals, pinning the CI half-width to ± 1.8 pp. Head-to-head comparisons use candidate as row and opponent as column.

Table 2. Tournament matrix: row bot win% vs. column bot, $n = 3000$ deals. WANN vs. Elite: 52.1% ± 1.8% (95% CI [50.3, 53.9]); card points 60.2 vs. 59.8.

	Random	**OldHeur**	**Elite**	**WANN**
RandomBot	50.0	10.8	4.7	5.1
OldHeuristicBot	89.2	50.0	32.5	32.3
EliteHeuristicBot	95.3	67.5	50.0	47.9
WANN	95.0	67.7	**52.1**	50.0

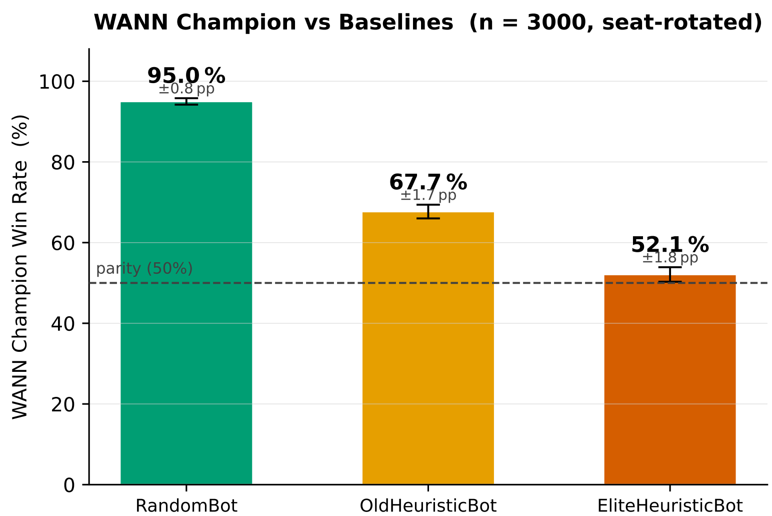

Fig. 5. Visualization of tournament results. The WANN's edge over Elite is small but statistically significant at $n = 3000$.

6.2 Tournament Results

Table 2 presents the full round-robin. The WANN beats Elite **52.1% ± 1.8%** ($n = 3000$), 95% CI $[50.3, 53.9]$, a statistically significant win excluding 50%. The card-point differential is 60.2 vs. 59.8. Against OldHeuristic the WANN achieves 67.7% (identical to Elite's 67.5%), and against RandomBot 95.0%.

6.3 From Prior Failure to the Final Champion

As introduced in Section 1, the pre-diagnosis champion reached only **30.2% vs. Elite** (46.3% vs. OldHeuristic). A fast training-free harness gave the collapse a concrete signature: EFFICIENT_WIN fired ~80% of the time, EQUITY_BUILDER was never used, and its output node was structurally unreachable from any input. Under the old resolver the three intents scored just 20/37/25% vs. Elite individually, so self-play atrophied the weaker two.

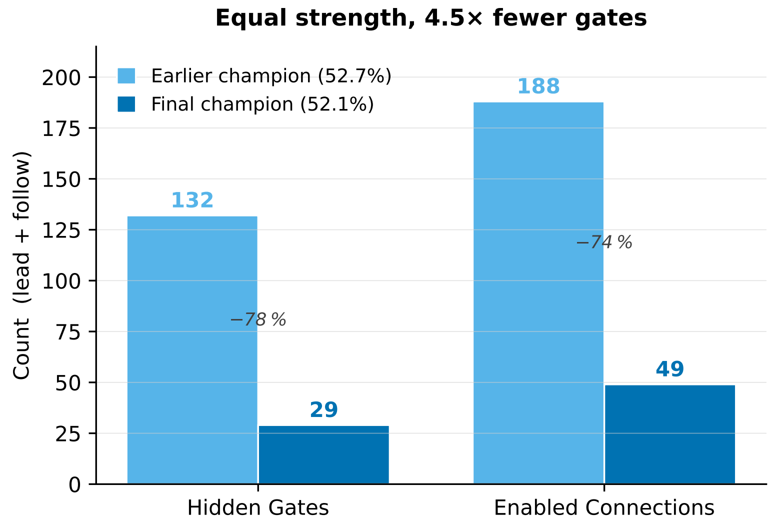

Fig. 6. Complexity comparison: the earlier alpha-beta-teacher champion (132 hidden gates) against the final rollout-teacher champion (29 gates). Equal playing strength, 4.5× simpler.

The **styled resolver** (Section 3.4) raised each pure intent to ≈Elite (48/47/46%), removing the collapse incentive. Combined with best-of-three-intent labelling, this produced the earlier (alpha-beta-teacher) champion at **52.7% ± 1.8%**, the first Elite-beating result. The **rollout teacher** then replaced the myopic alpha-beta dataset with supra-Elite labels (62% vs. Elite). The retrained final champion achieved a statistically identical **52.1% ± 1.8%** but with **4.5× fewer hidden gates** (29 vs. 132) and 3.8× fewer connections (49 vs. 188).

6.4 Diagnostic Ceilings

We established three quantitative ceilings: (i) **3-intent oracle** (perfect selection): **58%** vs. Elite, the absolute cap under the current vocabulary and resolver; (ii) **RandomForest on belief features**: predicting EFFICIENT_WIN vs. EQUITY_BUILDER achieves **62.6% on lead, 70.8% on follow** (5-fold CV), showing the features carry signal (the WANN's realised F-value is ∼47%); (iii) **Rollout teacher**: 62.0% vs. Elite, confirming the teacher is genuinely supra-Elite, yet the champion caps at ∼52%: the gap is the resolver-floor headroom.

7 Discussion

7.1 The Resolver-Floor Ceiling

The most consequential finding is that **the styled resolver is simultaneously the enabler of victory and the strength ceiling**. By making each intent

≈Elite, the resolver eliminated the Phase-1 collapse basin that had trapped every prior run below 31%. But the same design makes the three intents nearly equivalent in fitness: the worst is ∼46% vs. Elite, the best ∼48%. There is little gradient to learn precise selection, and the network converges to the simplest policy the resolver floors to Elite.

The learned policy is near-constant: `EFFICIENT_WIN = 0` in both brains. The lead brain chooses between "press-when-ahead" (MAX_FORCE) and "build-voids-when-trump-poor" (EQUITY_BUILDER). The follow brain gates equity on a cascade: boss-detection → opponent-void awareness → trump adequacy → threshold, while MAX_FORCE is simply $2 \cdot \text{Game\_Pts\_Remaining}$, a game-phase toggle (force early, concede late). The rules *are* strategically sensible, but their simplicity reflects the weakness of the selection signal, not a minimal description of optimal play.

Did we just learn Elite's IF/THEN? Substantially yes. The action space is Elite-floored; the WANN's job is to choose *when to deviate*, and it learned two deviations (equity-when-trump-poor, force-when-ahead). The network **never outputs EFFICIENT_WIN**: it always applies one of the two overlays. The WANN did not rediscover Elite from scratch; it learned a small situational overlay on a strong fixed base, and that overlay turns a 50% tie into 52%.

7.2 Limitations

Three-intent bottleneck. With 3 intents each ≈Elite, headroom above Elite is ≈8 pp (58% oracle cap). More differentiated intents would raise this. **Lead-brain bootstrap.** PFS-NEAT from zero connections can fail in the lead population (the first 1–2 connections cannot beat chance); production at pop=1000 bootstrapped fine. **Single game.** Sueca-specific design choices (35 features, lead/follow split) are plausible but untested for other trick-taking games. **Elite ≈ fast-policy ceiling.** The WANN's +2% edge is consistent with Elite being near the one-ply heuristic ceiling.

7.3 Open Question: Fog of War

Can a tiny logic circuit learn to *infer opponent cards* from assumed-optimal play using only public information? The belief state encodes voids and depletion but not explicit opponent-hand hypotheses (e.g. "partner probably holds the diamond Ace"). Humans routinely make such inferences; a WANN with richer inference features might do the same, a path to break the 58% oracle cap via better intent discrimination.

8 Conclusion and Future Work

We evolved an interpretable, Elite-beating Sueca agent with a verified pipeline from weight-agnostic logic networks to human-readable IF/THEN rules. The final champion compiles to 5 follow-brain logic gates at depth 5, a 4.5× compression over the earlier champion at equal strength (52.1% vs. Elite). The keys were a styled resolver that raised each intent's floor to Elite and a rollout teacher whose supra-Elite labels produced a simpler, more interpretable champion without improving strength.

Future directions: (1) **richer intents** to raise the 58% oracle cap; (2) **opponent-inference belief features** encoding hypotheses about unseen hands; (3) **knowledge distillation** into a smaller student for the full strength–interpretability Pareto frontier; (4) **connection pruning** of the genome beyond post-hoc rule folding.

The core finding, that a stronger teacher produces a *simpler* model at the same strength rather than a stronger one when the action space is floor-bounded, is likely to recur in neurosymbolic systems where a strong hand-authored base policy constrains the learned overlay: teacher quality governs interpretability, not performance.

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A Belief State Layout (35 features)

#	Feature	Description
0	Has_Led_Suit	Holds at least one card of the led suit
1	Has_Trump	Holds at least one trump card
2	Led_Suit_Count	Cards held in led suit / 10.0
3	Trump_Count	Trumps held / 10.0
4	Hand_Point_Density	Points in hand / points remaining in game
5	Am_I_Leading	First to play in the trick
6	Am_I_Last_To_Play	Fourth to play in the trick
7	Is_Partner_Winning	Partner currently winning the trick
8	Trick_Point_Value	Points on the table / 44.0
9	Has_Trick_Been_Cut	Trump played when led suit $\neq$ trump
10	Partner_Void_Led	Partner known void in led suit
11	Partner_Void_Trump	Partner known void in trump
12	Any_Opp_Void_Led	Either opponent void in led suit
13	Any_Opp_Void_Trump	Either opponent void in trump
14	Led_Suit_Ace_Played	Led suit Ace already played
15	Led_Suit_7_Played	Led suit 7 (manilha) already played
16	Trump_Ace_Played	Trump Ace already played
17	Holds_Boss_Led	Holds the highest unplayed in led suit
18	Holds_Boss_Trump	Holds the highest unplayed in trump
19	Can_Beat_Winner	Any legal card can beat current winner
20	Min_Winning_Cost	Points of cheapest winning card / 11.0
21	Min_Sacrifice_Cost	Points of cheapest legal card / 11.0
22	Game_Pts_Remaining	Unplayed card points / 120.0
23	Trick_Number	Current trick index / 9.0
24	Trumps_Remaining	Unplayed trump cards / 10.0
25	Score_Delta	(our−opp pts + 120) / 240.0
26	My_Void_Count	Suits the player is void in / 3.0
27	Longest_Side_Suit	Max cards in non-trump, non-led suit / 10.0
28	Shortest_Side_Suit	Min cards in non-trump, non-led suit / 10.0
29	Side0_Depletion	Played cards of side-suit 0 / 10.0
30	Side1_Depletion	Played cards of side-suit 1 / 10.0
31	Side2_Depletion	Played cards of side-suit 2 / 10.0
32	Points_Secured_Us	Our team's secured game points / 120.0
33	Known_Void_Suits_Count	Suits where any player known void / 4.0
34	Depleted_Suits_Count	Fully-depleted suits / 4.0

All features are normalized to $[0, 1]$. Bool features are $\{0.0, 1.0\}$. The input order is fixed; evolution may connect to any subset.